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Lab Title: Creating an Amazon Elastic Container Service for Kubernetes (EKS) Cluster using eksctl

Introduction

Elastic Container Service for Kubernetes (EKS) Cluster just like any Platform as a Service is a fully managed Kubernetes service provided by AWS. In other words, the infrastructure in the hood is managed by AWS. For instance, clients will not be worried about the management of the control plane, Kubernetes API server and so on. EKS in conjunction with other AWS resources such as Identity and Access Management (IAM), Virtual Private Cloud (VPC), Application Loadbalancer and cloudwatch can be used to deploy an infrastructure on the AWS management console.

Just like any container orchestrating tool, EKS is responsible for managing containerized applications and suitable for production environments.

With the help of eksctl which is a command line tool for creating and managing Amazon Elastic Container Service for Kubernetes (EKS) cluster, cluster creation could be simplified. There is heavy lifting in the creation of VPC, Security group, integration of IAM with cloudwatch and other provisioning in the management console essential for a secure, scalable, and reliable EKS deployment.

Demo

In this lab, the service eksctl was used to create a cluster leveraging Amazon Elastic Container Service for Kubernetes (EKS) from AWS. It involved installing eksctl, configuring AWS credentials and creating an EKS cluster. The creation of the EKS cluster involved just a one line command with configurations such as the cluster name , the node group name , the region, the node type an the number of nodes.

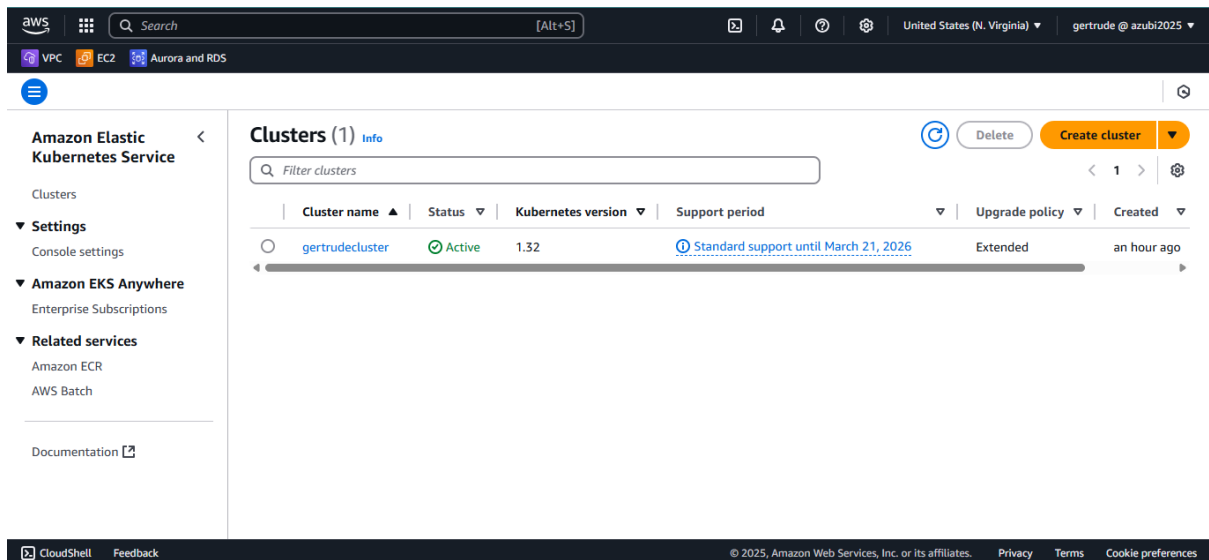
Below are screenshots capturing the processes involved in creating the EKS cluster using VS code terminal which was confirmed by referencing the AWS management console.

The screenshot shows the initial setup of an EKS cluster. The terminal output includes the following commands and their results:

```
PS C:\Users\gertr> choco install aws
PS C:\Users\gertr> eksctl version
0.207.0
PS C:\Users\gertr> aws configure
AWS Access Key ID [*****]:
AWS Secret Access Key [*****]:
Default region name [us-east-1]:
Default output format [json]:
PS C:\Users\gertr> eksctl create cluster --name gertrudecluster --nodegroup-name dnd1 --region us-east-1 --node-type t3.medium --nodes 2
2025-05-03 08:55:41 [i] eksctl version 0.207.0
2025-05-03 08:55:41 [i] using region us-east-1
2025-05-03 08:55:42 [i] setting availability zones to [us-east-1a us-east-1c]
2025-05-03 08:55:42 [i] subnets for us-east-1a - public:192.168.0.0/19 private:192.168.64.0/19
2025-05-03 08:55:42 [i] subnets for us-east-1c - public:192.168.32.0/19 private:192.168.96.0/19
2025-05-03 08:55:42 [i] nodegroup "dnd1" will use "" [AmazonLinux2/1.32]
2025-05-03 08:55:42 [i] using Kubernetes version 1.32
2025-05-03 08:55:42 [i] creating EKS cluster "gertrudecluster" in "us-east-1" region with managed nodes
2025-05-03 08:55:42 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2025-05-03 08:55:42 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=gertrudecluster'
2025-05-03 08:55:42 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "gertrudecluster" in "us-east-1"
2025-05-03 08:55:42 [i] CloudWatch logging will not be enabled for cluster "gertrudecluster" in "us-east-1"
2025-05-03 08:55:42 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=gertrudecluster'
2025-05-03 08:55:42 [i] default addons vpc-cni, kube-proxy, coredns, metrics-server were not specified, will install them as EKS addons
2025-05-03 08:55:42 [i]
2 sequential tasks: { create Cluster control plane "gertrudecluster",
2 sequential sub-tasks: {
2 sequential sub-tasks: {
1 task: { create addons },
wait for control plane to become ready,
},
create managed nodegroup "dnd1",
```

The screenshot shows the completion of the EKS cluster setup. The terminal output includes the following commands and their results:

```
2025-05-03 09:01:48 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-cluster"
2025-05-03 09:02:49 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-cluster"
2025-05-03 09:03:50 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-cluster"
2025-05-03 09:03:54 [i] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the required permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under "addon.PodIdentityAssociations", and run 'eksctl update addon'
2025-05-03 09:03:54 [i] creating addon: vpc-cni
2025-05-03 09:03:55 [i] successfully created addon: vpc-cni
2025-05-03 09:03:55 [i] creating addon: kube-proxy
2025-05-03 09:03:56 [i] successfully created addon: kube-proxy
2025-05-03 09:03:56 [i] creating addon: coredns
2025-05-03 09:03:57 [i] successfully created addon: coredns
2025-05-03 09:03:57 [i] creating addon: metrics-server
2025-05-03 09:03:58 [i] successfully created addon: metrics-server
2025-05-03 09:06:02 [i] building managed nodegroup stack "eksctl-gertrudecluster-nodegroup-dnd1"
2025-05-03 09:06:03 [i] deploying stack "eksctl-gertrudecluster-nodegroup-dnd1"
2025-05-03 09:06:03 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-nodegroup-dnd1"
2025-05-03 09:06:34 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-nodegroup-dnd1"
2025-05-03 09:07:13 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-nodegroup-dnd1"
2025-05-03 09:08:44 [i] waiting for CloudFormation stack "eksctl-gertrudecluster-nodegroup-dnd1"
2025-05-03 09:08:45 [i] waiting for the control plane to become ready
2025-05-03 09:08:46 [✓] saved kubeconfig as "C:\Users\gertr\kube\config"
2025-05-03 09:08:46 [i] no tasks
2025-05-03 09:08:46 [✓] all EKS cluster resources for "gertrudecluster" have been created
2025-05-03 09:08:46 [i] nodegroup "dnd1" has 2 node(s)
2025-05-03 09:08:46 [i] node "ip-192-168-18-31.ec2.internal" is ready
2025-05-03 09:08:47 [i] node "ip-192-168-18-31.ec2.internal" is ready
2025-05-03 09:08:47 [i] node "ip-192-168-38-39.ec2.internal" is ready
2025-05-03 09:08:47 [✓] created 1 managed nodegroup(s) in cluster "gertrudecluster"
2025-05-03 09:08:49 [i] kubectl command should work with "C:\Users\gertr\kube\config", try 'kubectl get nodes'
2025-05-03 09:08:49 [✓] EKS cluster "gertrudecluster" in "us-east-1" region is ready
PS C:\Users\gertr>
```



Overall, the `eksctl` was only needed to configure the cluster. The management of the EKS cluster which involved networking and provisioning of the control plane was the responsibility of AWS. This cluster could be used to orchestrate containerized applications allowing user to focus on the development of their application.